UNIT 2 · 9 hours

# Speed Control of a DC Shunt Motor (Armature and Field)



Transformers and Electrical Machines

**Basic Electrical and Electronics Engineering (EEE103)**
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
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# Learning objectives

- Derive the speed equation of a DC motor
- Explain armature resistance control
- Explain field (flux) control
- Compare the two methods and solve numericals

**Syllabus context**

DC Motor: principle of operation, back EMF and its significance, torque equation, characteristics and speed control (armature and field) of DC shunt motor.

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# Speed equation

$$N = (V - I_a R_a) / (K \varphi)$$

*K = ZP/60A (constant)*

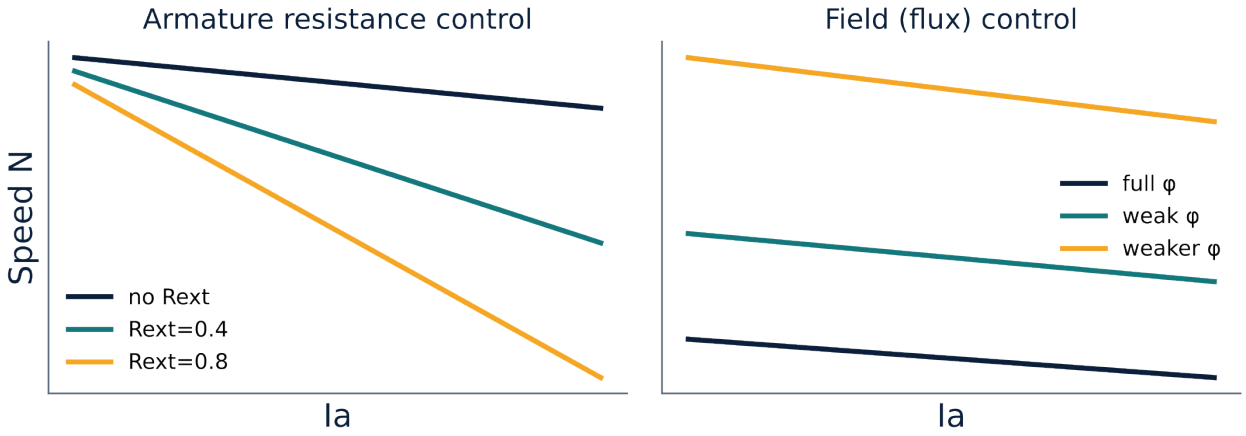
$$N \propto E_b / \varphi$$

*approximately $N \propto 1/\varphi$*

- Speed can be varied by changing: V, Ra (armature circuit), or φ (field).
- Syllabus methods: armature resistance control and field flux control.

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# Armature resistance control




- Variable resistance Rext in series with the armature.
- Field is across the supply, so flux is unchanged.
- Ia Ra drop increases → Eb falls → speed falls.
- Speed can be reduced below normal only.

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# Armature resistance versus field control

## Armature resistance control

- Speed below rated value
- Speed falls with load (poor regulation)
- Large $I^2R$ loss in Rext, so inefficient
- Used for short-time duty or small motors

## Field (flux) control

- Speed above rated value
- Rheostat in shunt field circuit; weaker flux gives higher speed
- Small field current, so very low loss
- Limit: very weak field harms commutation

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# Field control in detail

- Rheostat Rc in series with the shunt field.
- Increase Rc → Ish falls → φ falls → N rises ($N \propto 1/\varphi$).
- Armature current rises when flux is reduced (torque requirement constant: $T_a \propto \varphi I_a$).
- Speed range is limited, typically up to 2:1.
- Field-open danger: φ residual only → speed runaway; use field failure protection.

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# Numerical: armature resistance control

**Problem** A 220 V shunt motor, Ra = 0.5 Ω, runs at 1000 rpm with Ia = 20 A. A 2.5 Ω resistance is inserted in series with the armature, load torque constant. Find the new speed.

- Constant torque, constant φ → Ia = 20 A unchanged
- Eb1 = 220 − 20 × 0.5 = 210 V
- Eb2 = 220 − 20 × (0.5 + 2.5) = 160 V
- N2 = N1 × Eb2/Eb1 = 1000 × 160/210

**Answer: N2 = 762 rpm**

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# Numerical: field control

**Problem** Same motor. The field flux is reduced by 10 % while Eb stays nearly the same. Estimate the new speed.

- $N \propto Eb/\varphi$;  $Eb \approx$ constant
- $N2 = N1 \times (\varphi1/\varphi2) = 1000/0.9$

**Answer:  N2 ≈ 1111 rpm**

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# Key takeaways

- $N = (V - I_a R_a)/(K\varphi)$
- Armature resistance control lowers speed; inefficient
- Field control raises speed; efficient, small field current
- Never open the shunt field of a running motor

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# Practice questions

**Q1.** Explain how the speed of a DC shunt motor is varied by field control.

**Q2.** Why is armature resistance control inefficient?

**Q3.** A shunt motor at 1200 rpm: flux reduced by 20 %. Estimate new speed.

**Q4.** Compare the two methods in a table.